Property $\mathcal{A}$ does not force frequent hypercyclicity

Literature-implied status packet for arXiv:1008.4017

Status

This is a `literature_implied_answer` packet. It records that Menet's later chaotic non-frequently-hypercyclic operator gives a negative answer to a stronger question posed by Costakis–Parissis in *Szemeredi's theorem, frequent hypercyclicity and multiple recurrence*, arXiv:1008.4017 [1].

Source question

In Section 2, Costakis–Parissis define property $\mathcal{A}$: a sequence $(T_n)_{n\in\mathbb{N}}$ has property $\mathcal{A}$ if for every open set $U \subset X$ there is $x \in X$ such that

$$\overline{\mathrm{dens}}\{n \in \mathbb{N} : T_n x \in U\} > 0.$$

For a single operator T, this means that $(T^n)_{n\in\mathbb{N}}$ has property $\mathcal{A}$.

In the final "Further Questions" section they write:

> It is easy to see that every chaotic operator T has property $\mathcal{A}$. A well known question asks whether every chaotic operator is frequently hypercyclic; see for example [2]. An even stronger question is thus whether every hypercyclic operator that has property $\mathcal{A}$ is frequently hypercyclic.

Supporting answer

Quentin Menet's later paper *Linear chaos and frequent hypercyclicity*, arXiv:1410.7173 [3], answers the embedded chaotic-operator problem negatively. Theorem 1.2 states that there exists a chaotic operator T on ℓ^1 which is neither $\mathcal{U}$-frequently hypercyclic nor distributionally chaotic; in particular, T is chaotic and not frequently hypercyclic.

Combining these two facts gives the advertised counterexample to the stronger source question. Menet's operator is chaotic, so it is hypercyclic. By the Costakis–Parissis observation, it has property $\mathcal{A}$. But Theorem 1.2 of Menet shows that it is not frequently hypercyclic. Therefore

$$\text{hypercyclic} + \mathcal{A} \not\Rightarrow \text{ frequently hypercyclic.}$$

Scope limitations

This packet is not a new construction from the run; it is a short later-literature implication. It settles only the frequent-hypercyclicity question above. The subsequent Costakis–Parissis question asking whether there exists an operator with property $\mathcal{A}$ which is not $\mathcal{U}$-frequently recurrent is not settled here, because Menet's Theorem 1.2 concerns $\mathcal{U}$-frequent hypercyclicity rather than $\mathcal{U}$-frequent recurrence.

References

[1] G. Costakis and I. Parissis, *Szemeredi's theorem, frequent hypercyclicity and multiple recurrence*, arXiv:1008.4017.

[2] F. Bayart and E. Matheron, *Dynamics of linear operators*, Cambridge University Press, 2009.

[3] Q. Menet, *Linear chaos and frequent hypercyclicity*, arXiv:1410.7173.